

OPEN ENDED EXPERIMENT – 2

FUEL LEAKAGE DETECTION AND ENGINE PROTECTION SYSTEM

Aim of the Experiment

To detect fuel-related vapors using an MQ-3 gas sensor interfaced with the STM32 microcontroller and to automatically stop the DC motor (simulated engine) while activating a warning LED when vapor concentration exceeds a predefined safety threshold.

List of Equipment / Components Required

Sl. No.	Component / Hardware	Specification	Quantity
1	STM32 Blue Pill Development Board	STM32F103C8T6	1
2	MQ-3 Gas Sensor Module	Alcohol / Vapor Sensor	1
3	DC Motor	5 V–12 V DC Motor	1
4	Green LED	Status Indicator	1
5	Red LED	Warning Indicator	1
6	Resistors	330 Ω	2
7	Connecting Wires	Jumper Wires	As Required

Theoretical Background

The STM32 Blue Pill microcontroller is widely used in embedded and automotive monitoring applications because of its high processing speed, Analog-to-Digital Converter (ADC) capability, and GPIO control features. In this project, an MQ-3 gas sensor is used to detect fuel-related vapors, while a DC motor represents the vehicle engine. A green LED indicates safe operating conditions and a red LED indicates a hazardous condition.

Fuel leakage is one of the major safety concerns in automotive systems. Leakage from fuel pipes, loose connections, damaged hoses, or accidental impacts can release fuel vapors into the surrounding environment. These vapors are highly flammable and may lead to fire hazards when exposed to high temperatures or electrical sparks.

The MQ-3 sensor is sensitive to alcohol and other volatile flammable vapors. In this prototype, fuel leakage is simulated using alcohol-based vapors such as sanitizer or spirit. The sensor produces an analog voltage proportional to the concentration of vapors present in the air. This analog signal is continuously monitored by the STM32 microcontroller through its ADC module.

During normal conditions, the vapor concentration remains below a predefined threshold. The STM32 interprets this as safe operation and keeps the DC motor running while turning on the green LED. When the vapor concentration exceeds the threshold value, the STM32 identifies a potential fuel leakage condition. It immediately stops the DC motor to simulate engine shutdown, turns off the green LED, and activates the red LED as a warning indication.

This project demonstrates the practical implementation of sensor interfacing, ADC operation, threshold-based decision making, GPIO control, and automotive safety monitoring using STM32. The system provides an effective method for detecting hazardous fuel vapor conditions and initiating protective actions before dangerous situations occur.

Interfaces Between Components

Sl. No.	Component / Module	Input	Output	Function
1	MQ-3 Gas Sensor	Vapor Concentration	Analog Voltage	Gas Detection
2	STM32 ADC	Analog Voltage	Digital ADC Value	Analog-to-Digital Conversion
3	STM32 GPIO	ADC Digital Value	Motor / LED Signal	Threshold Decision & Control
4	DC Motor	GPIO Drive Signal	Mechanical Output	Engine Simulation
5	Green LED	GPIO Signal (Normal)	Green Light	Safe Status Indicator
6	Red LED	GPIO Signal (Alarm)	Red Light	Warning Indicator

Simulink Model

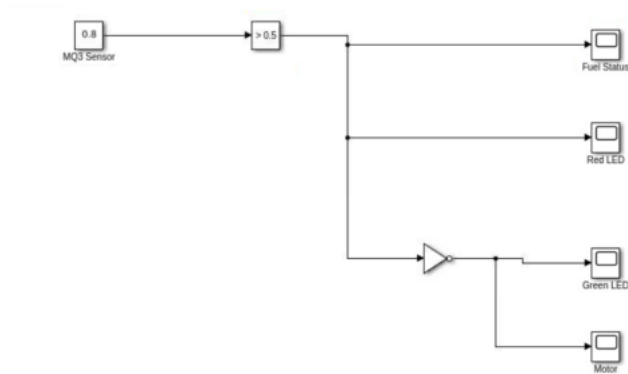


Figure 26: Simulink Model – Fuel Leakage Detection and Engine Protection System

Breadboard Implementation

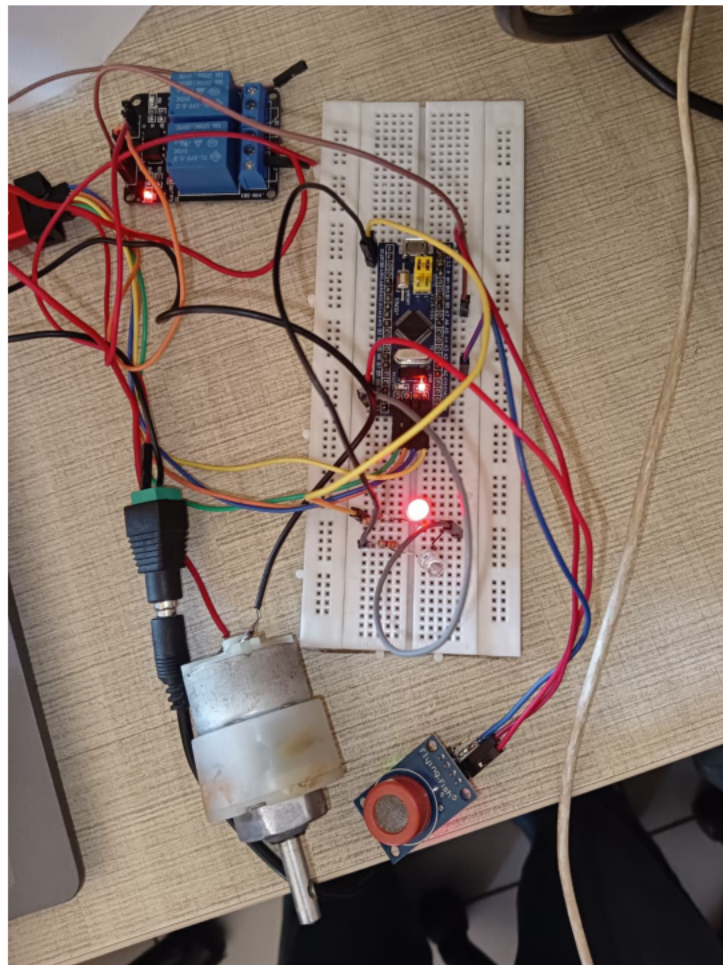


Figure 27: Breadboard Implementation – Fuel Leakage Detection and Engine Protection System

Result and Conclusion

The STM32 Blue Pill successfully monitored the MQ-3 sensor output and detected the presence of fuel-related vapors. During normal operating conditions, the vapor concentration remained below the threshold value, causing the green LED to remain on and the DC motor to continue running, indicating safe vehicle operation.

When alcohol-based vapors were introduced near the MQ-3 sensor to simulate fuel leakage, the sensor output increased significantly. Once the detected vapor concentration exceeded the predefined threshold, the STM32 immediately activated the protection mechanism: the green LED was turned off, the red LED was turned on, and the DC motor was stopped to represent engine shutdown.

The project successfully demonstrated a fuel leakage detection and engine protection system using STM32. The system effectively combined sensor monitoring, ADC processing, threshold comparison, and actuator control to detect hazardous vapor conditions and initiate protective action. The results confirm that STM32-based monitoring systems can be used for automotive safety applications where early detection of fuel leakage is essential for preventing accidents and improving vehicle safety.